VisionGuard AI — UI/UX Design Document

1. Design direction

VisionGuard AI should feel like a calm, modern operations dashboard: clean, spacious, professional, and easy to use under pressure. The interface should present complex live-camera and AI information through a modular grid of soft white cards, subtle shadows, fine gray dividers, and restrained color accents.

The design should avoid a cluttered “security control room” appearance. Instead, it should feel polished, approachable, and focused—suitable for restaurant managers, sports stewards, security teams, factory supervisors, and business owners.

2. Design principles

- **Calm first:** Important alerts stand out, but the interface should not feel alarming by default.
- **Clear hierarchy:** Live status, active faults, and required actions are always visible.
- **Modular layout:** Every function lives in a rounded, independent card.
- **Low cognitive load:** Use simple labels, icons, and consistent spacing.
- **Fast scanning:** Operators should understand site status in under five seconds.
- **Accessible contrast:** Text must remain readable on white, warm-gray, and accent-colored surfaces.
- **Responsive:** Optimize first for desktop, then tablet and mobile.

3. Visual style

Layout

- Pale warm-gray page background.
- One centered main application panel with large rounded corners.
- Subtle drop shadow around the main panel to create gentle depth.
- Generous whitespace between cards and sections.
- A precise modular grid: 12-column desktop grid, 8-column tablet grid, single-column mobile grid.

Cards

- White or very light warm-white backgrounds.
- Border radius: 16–20 px for large cards, 12–14 px for small cards.
- Soft shadow: low opacity, large blur, no harsh edges.
- Fine light-gray divider lines inside cards where needed.
- No heavy borders; use spacing and shadow to separate content.

Typography

- Minimalist modern sans-serif typeface.
- Recommended fonts: Inter, Plus Jakarta Sans, Manrope, or IBM Plex Sans.
- Charcoal for primary text.
- Medium gray for secondary text.
- Light gray for labels, timestamps, and metadata.
- Sentence case for most interface labels.
- Numbers should use tabular figures for counts, confidence scores, timestamps, and camera IDs.

Type scale:

Style	Size	Weight	Use
Display	32–40 px	600–700	Page greeting or major KPI
H1	24–28 px	600	Page titles
H2	18–20 px	600	Card titles
H3	15–16 px	600	Section labels
Body	14 px	400–500	General content
Small	12–13 px	400–500	Metadata, timestamps, helper text
Micro	10–11 px	500–600	Badges and tiny labels

Color palette

Role	Color	Use
Page background	#F4F2EF	App background
Surface	#FFFFFF	Cards, panels, modals
Charcoal	#232323	Primary text, icons
Warm gray	#6E6A66	Secondary text
Light gray	#D9D5D0	Dividers, inactive elements

Role	Color	Use
Coral	#F26B5B	Primary action, active alert, selected state
Soft peach	#FAD9CE	Chart fills, soft highlights, alert backgrounds
Blush	#F7E3DC	Secondary chart layer, subtle backgrounds
Success green	#3E9B6C	Online cameras, resolved events
Warning amber	#E0A63C	Medium-severity alerts
Danger red	#D64545	Critical alerts only

Coral is the primary brand/action color. Red should be reserved for critical safety or security incidents so it retains meaning.

Iconography

- Thin, monochrome, rounded-line icons.
- Consistent 1.5–2 px stroke weight.
- Icons should be simple and recognizable: camera, screen, alert, user, chart, shield, football, restaurant, factory, settings.
- Use color only when an icon represents state or severity.

Imagery

- No large human illustrations or stock photography.
- Camera thumbnails and small avatar thumbnails are acceptable.
- Use real camera frames only inside clearly defined video cards.
- Keep the interface vector-like, crisp, and uncluttered.

4. App structure

Left sidebar

A slim, fixed navigation sidebar on desktop.

Contents:

- Brand mark at the top.
- Primary navigation:
  - Live Operations
  - Fault Center
  - Cameras & Screens
  - Rules & Actions
  - Training Studio

    - Models
    - Reports
    - Settings
- Site selector near the bottom or top.
- User avatar and account menu.

The selected item uses a warm coral highlight, soft peach background, or coral left indicator. Inactive items use charcoal or warm-gray icons and text.

Top bar

A slim top bar inside the main panel.

Contents:

- Current site name on the left.
- Global search for cameras, events, rules, and models.
- Notification bell with unread-alert count.
- Date/time badge.
- User avatar and name.

Main workspace

The main area uses a spacious card grid. The dashboard should always answer:

1. Is everything normal?
2. What needs attention now?
3. Which camera or screen is involved?
4. What action should be taken?

5. Dashboard layout

Top greeting area

A large, calm greeting anchors the upper center or upper left:

```
Good morning, Christian
Your venues are being monitored. 2 events need review.
```

Beside it:

- Date badge: `Wed, 04 Nov 2026`
- Coral primary button: `Add Camera` or `New Rule`
- Optional secondary button: `Run System Check`

Status strip

A horizontal row of compact KPI cards:

KPI	Example
Cameras online	7 / 8
Active faults	2
Events today	18
Critical alerts	0
False positives this week	6
System health	Healthy

Each KPI card contains:

- Small icon.
- Label.
- Large number.
- Small trend indicator or status dot.

Live operations card

A large card showing the camera grid.

- Each camera appears as a rounded video thumbnail.
- Offline cameras show a muted gray placeholder with a “Camera offline” label.
- Active detections show a thin coral outline or small status badge.
- Each thumbnail includes:
  - Camera name.
  - Site/location.
  - Live indicator.
  - Last event time.
  - Small expand/full-screen icon.

Recommended grid:

- Desktop: 2–4 camera tiles per row.
- Tablet: 2 per row.
- Mobile: 1 per row.

Fault Center card

A high-priority card listing current and recent events.

Each event row contains:

- Severity dot: coral/red for critical, amber for medium, gray for low.
- Fault type: “Pitch invasion,” “Queue buildup,” “Possible leak.”
- Camera name and site.
- Timestamp.
- Confidence score.
- Snapshot thumbnail.
- Status: Open, Acknowledged, Resolved, False positive.
- Action buttons: View, Acknowledge, Resolve, Mark false positive.

Critical events should appear at the top with a soft peach or coral-tinted background—not a full red alarm background.

Analytics card

A compact analytics panel with:

- Events per day bar chart.
- Fault-type breakdown.
- False-positive trend.
- Camera activity heat strip.

Use soft peach and blush chart fills with charcoal axis labels. Avoid heavy gridlines.

Model health card

Shows the active AI model:

- Model name.
- Version.
- Status: Active.
- Accuracy/mAP.
- Last trained date.
- Buttons: Test model, View metrics, Rollback.

Actions card

Displays automated actions and their status:

- Telegram alert sent.
- Email sent.
- Webhook called.
- Siren triggered.
- PLC stop requested.

Each action includes a status indicator: Pending, Sent, Failed, Acknowledged.

6. Key screens

1. Live Operations

Purpose: monitor all cameras and screens.

Layout:

- Status strip.
- Large camera grid.
- Right-side active-alert panel.
- Bottom recent-events strip.

2. Fault Center

Purpose: review and manage events.

Layout:

- Filters: site, camera, severity, status, date, fault type.
- Event table/cards.
- Event detail drawer on the right.
- Evidence viewer with snapshot, clip, timeline, and detection boxes.

3. Cameras & Screens

Purpose: manage sources.

Layout:

- Camera cards.
- Add-camera button.
- Connection status.
- Resolution, FPS, active model.

- Zone editor preview.

4. Rules & Actions

Purpose: configure automated decisions.

Layout:

- Rule list.
- Rule builder panel.
- Trigger, condition, persistence, cooldown, severity, and action sections.
- Test-rule button.

5. Training Studio

Purpose: create and improve AI models.

Layout:

- Dataset upload card.
- Class manager.
- Labeling progress.
- Training progress with loss/mAP charts.
- Model comparison.
- Activate/Rollback controls.

6. Reports

Purpose: show business value.

Layout:

- Date-range selector.
- Events by site.
- Events by fault type.
- Response-time chart.
- False-positive rate.
- Export CSV/PDF button.

7. Component library

Buttons

Type	Style
Primary	Coral background, white text, 10–12 px radius
Secondary	White background, light-gray border, charcoal text
Ghost	Transparent background, warm-gray text
Danger	Red background, white text, used only for critical actions
Disabled	Light-gray background, muted text

Badges

- `Online`: soft green background, dark-green text.
- `Offline`: light-gray background, warm-gray text.
- `Critical`: soft red background, dark-red text.
- `High`: soft coral background, dark-coral text.
- `Medium`: soft amber background, dark-amber text.
- `Low`: light-gray background, charcoal text.
- `Active model`: soft peach background, charcoal text.

Alerts

- Use cards, not full-screen popups.
- Critical alerts may use a coral/red left border and soft tinted background.
- Include snapshot, fault name, camera, timestamp, confidence, and action buttons.
- Provide “Acknowledge” and “View evidence” as primary actions.

Forms

- Rounded input fields with light-gray borders.
- Clear labels above fields.
- Helper text below fields.
- Validation messages in red only after user interaction.
- Use toggles for enabling cameras, rules, and alert channels.

Tables

- White background.
- Light-gray row dividers.
- No heavy zebra striping.
- Sticky header.

- Compact rows with 12–14 px text.
- Status and severity shown as badges, not plain text.

Charts

- Minimal axis lines.
- Light-gray gridlines.
- Soft peach/coral fills.
- Charcoal labels.
- No 3D effects.
- Use bar charts for event counts, line charts for trends, and donut/gauge charts for status distribution.

8. Interaction design

States

Every interactive element needs:

- Default.
- Hover.
- Focus.
- Active/selected.
- Disabled.
- Loading.
- Error.

Micro-interactions

- Smooth 150–250 ms transitions.
- Subtle card hover elevation.
- Animated status dots for live cameras.
- Progress bars for training jobs.
- Skeleton loaders for camera grids and event tables.
- Toast notifications for successful actions.

Empty states

Use friendly, concise empty states:

```
No cameras connected yet.
Add your first camera to begin monitoring.
[Add Camera]
```

```
No active faults.
Your site is operating normally.
```

```
No trained models yet.
Upload a dataset to create your first custom detector.
[Start Training]
```

9. Accessibility

- Minimum text contrast: 4.5:1 for normal text.
- Minimum touch target: 44 x 44 px.
- Keyboard navigation for all main actions.
- Visible focus rings.
- Do not rely on color alone; pair color with icons, labels, or text.
- Provide alt text for camera thumbnails and event snapshots.
- Support reduced-motion preferences.

10. Responsive behavior

Breakpoint	Layout
Desktop 1440 px+	Full sidebar, 12-column grid, multi-camera grid
Desktop 1024–1439 px	Collapsible sidebar, 8-column grid
Tablet 768–1023 px	Icon-only sidebar, 2-column cards
Mobile below 768 px	Bottom navigation or hamburger menu, single-column cards

Mobile priority:

1. Active alerts.
2. Camera status.
3. Event review.
4. Acknowledge/resolve actions.
5. Settings.

11. Motion and feedback

- Use subtle fade/slide for drawers and modals.
- Animate only meaningful changes: new alert, training progress, camera status.
- Avoid continuous distracting animation.
- Live camera feeds should not have decorative effects.
- Critical alert appearance may use one brief emphasis animation, then remain static.

12. Design tokens

```
:root {
  --bg-page: #F4F2EF;
  --surface: #FFFFFF;
  --text-primary: #232323;
  --text-secondary: #6E6A66;
  --text-muted: #9C968F;
  --divider: #E7E3DE;
  --brand: #F26B5B;
  --brand-soft: #FAD9CE;
  --brand-blush: #F7E3DC;
  --success: #3E9B6C;
  --warning: #E0A63C;
  --danger: #D64545;

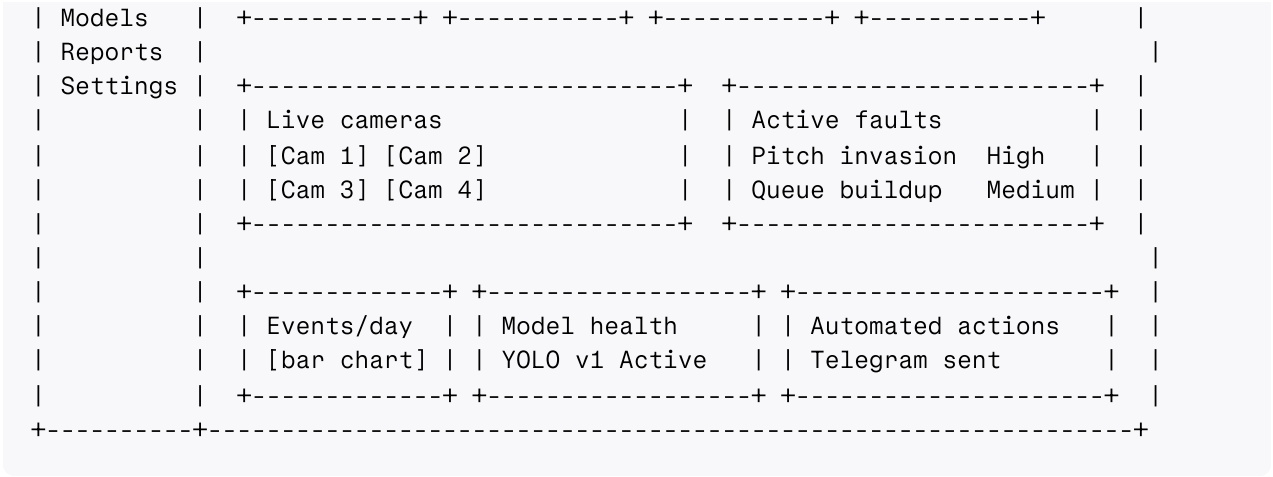
  --radius-lg: 20px;
  --radius-md: 14px;
  --radius-sm: 10px;

  --shadow-card: 0 10px 30px rgba(35, 35, 35, 0.06);
  --shadow-float: 0 18px 45px rgba(35, 35, 35, 0.10);

  --space-1: 4px;
  --space-2: 8px;
  --space-3: 12px;
  --space-4: 16px;
  --space-5: 24px;
  --space-6: 32px;
  --space-7: 48px;
}
```

13. Example dashboard wireframe

```
+-------------------------------------------------------------------------+
| [Logo] VisionGuard AI        [Search cameras, events, rules]   [Bell] [Avatar] |
+----------+--------------------------------------------------------------+
|          |  Good morning, Christian                                      |
| Live     |  2 events need review.          [Wed, 04 Nov] [Add Camera]    |
| Faults   |                                                               |
| Cameras  |  +-----------+ +-----------+ +-----------+ +-----------+      |
| Rules    |  | Cameras   | | Active    | | Events    | | Critical  |      |
| Training |  | 7 / 8     | | faults 2  | | today 18  | | alerts 0  |      |
```



14. Final design decision

Use a **white, rounded, modular dashboard on a pale warm-gray background**, with charcoal text, warm-gray secondary text, coral primary actions, and soft peach/blush data visualizations. The sidebar should be slim and icon-led; the main workspace should prioritize live cameras, active faults, evidence review, model health, and automated actions.

The result should feel like a polished, modern AI operations platform—not a financial app, not a generic admin panel, and not an overwhelming security console.